

DBSE&DBD

(2520030234) WEEK-2

```
Command Prompt - mysql -u X + v
Microsoft Windows [Version 10.0.26200.8973]
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C:\Users\T.Karthik>cd C:\Program Files\MySQL\MySQL Server 8.0\bin

C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u root -p
Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 14
Server version: 8.0.46 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE library_db;
Query OK, 1 row affected (0.01 sec)

mysql> USE library_db;
Database changed
mysql> CREATE TABLE Books (
    ->     book_id INT PRIMARY KEY,
    ->     title VARCHAR(100) NOT NULL,
    ->     isbn VARCHAR(20) UNIQUE,
    ->     published_year INT CHECK (published_year <= 2027)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO Books
    -> (book_id,title,isbn,published_year)
    -> VALUES
    -> (1,'The Great Gatsby','9780743273565',1925),
    -> (2,'To Kill a Mockingbird','9780061120084',1960),
    -> (3,'1984','9780451524935',1949);
Query OK, 3 rows affected (0.02 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM Books;
```

book_id	title	isbn	published_year
1	The Great Gatsby	9780743273565	1925
2	To Kill a Mockingbird	9780061120084	1960
3	1984	9780451524935	1949

```
3 rows in set (0.00 sec)
```



```
mysql> CREATE TABLE Members(
  ->     member_id INT PRIMARY KEY,
  ->     full_name VARCHAR(100),
  ->     email VARCHAR(100) UNIQUE
  -> );
```

```
Query OK, 0 rows affected (0.04 sec)
```



```
mysql> INSERT INTO Members
  -> (member_id,full_name,email)
  -> VALUES
  -> (101,'John Smith','john.smith@email.com'),
  -> (102,'Emma Wilson','emma.wilson@email.com'),
  -> (103,'Michael Brown','michael.brown@email.com');
```

```
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```



```
mysql> SELECT * FROM Members;
```

member_id	full_name	email
101	John Smith	john.smith@email.com
102	Emma Wilson	emma.wilson@email.com
103	Michael Brown	michael.brown@email.com

```
3 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE Loans(
->     loan_id INT PRIMARY KEY,
->     member_id INT,
->     book_id INT,
->     loan_date DATE,
->     FOREIGN KEY(member_id)
->         REFERENCES Members(member_id),
->     FOREIGN KEY(book_id)
->         REFERENCES Books(book_id)
-> );
```

Query OK, 0 rows affected (0.07 sec)

```
mysql> INSERT INTO Loans
-> (loan_id,member_id,book_id,loan_date)
-> VALUES
-> (1,101,1,'2025-01-05'),
-> (2,102,2,'2025-01-08'),
-> (3,103,3,'2025-01-10'),
-> (4,101,2,'2025-02-01'),
-> (5,102,1,'2025-02-05'),
-> (6,103,2,'2025-02-12'),
-> (7,101,3,'2025-03-01'),
-> (8,102,3,'2025-03-07'),
-> (9,103,1,'2025-03-15'),
-> (10,101,1,'2025-04-01');
```

Query OK, 10 rows affected (0.03 sec)

Records: 10 Duplicates: 0 Warnings: 0

```
mysql> SELECT * FROM Loans;
```

loan_id	member_id	book_id	loan_date
1	101	1	2025-01-05
2	102	2	2025-01-08
3	103	3	2025-01-10
4	101	2	2025-02-01
5	102	1	2025-02-05
6	103	2	2025-02-12
7	101	3	2025-03-01
8	102	3	2025-03-07
9	103	1	2025-03-15
10	101	1	2025-04-01

10 rows in set (0.00 sec)

```
mysql> SELECT m.full_name As Member_Name, b.title As Book_Title
-> FROM Loans l
-> INNER JOIN Members m ON l.member_id = m.member_id
-> INNER JOIN Books b ON l.book_id = b.book_id;
```

Member_Name	Book_Title
John Smith	The Great Gatsby
John Smith	To Kill a Mockingbird
John Smith	1984
John Smith	The Great Gatsby
Emma Wilson	To Kill a Mockingbird
Emma Wilson	The Great Gatsby
Emma Wilson	1984
Michael Brown	1984
Michael Brown	To Kill a Mockingbird
Michael Brown	The Great Gatsby

10 rows in set (0.00 sec)

```
mysql> SELECT published_year, COUNT(book_id) AS Total_Books
-> FROM Books
-> GROUP BY published_year
-> ORDER BY published_year;
```

published_year	Total_Books
1925	1
1949	1
1960	1

3 rows in set (0.01 sec)

```

mysql> CREATE TABLE Donation_History(
->     donation_id INT PRIMARY KEY,
->     book_id INT,
->     donor_name VARCHAR(100),
->     donation_date DATE,
->     FOREIGN KEY(book_id)
->     REFERENCES Books(book_id)
-> );
Query OK, 0 rows affected (0.04 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> INSERT INTO Books
-> (book_id,title,isbn,published_year)
-> VALUES
-> (4,'Animal Farm','9780451526342',1945);
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO Donation_History
-> (donation_id,book_id,donor_name,donation_date)
-> VALUES
-> (1,4,'Raj Kumar',CURDATE());
Query OK, 1 row affected (0.01 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

mysql> CREATE INDEX idx_books_isbn
-> ON Books(isbn);
Query OK, 0 rows affected (0.05 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> SELECT *
-> FROM Books
-> WHERE isbn='9780451524935';
+-----+-----+-----+-----+
| book_id | title | isbn          | published_year |
+-----+-----+-----+-----+
|      3 | 1984  | 9780451524935 |      1949      |
+-----+-----+-----+-----+
1 row in set (0.01 sec)

```